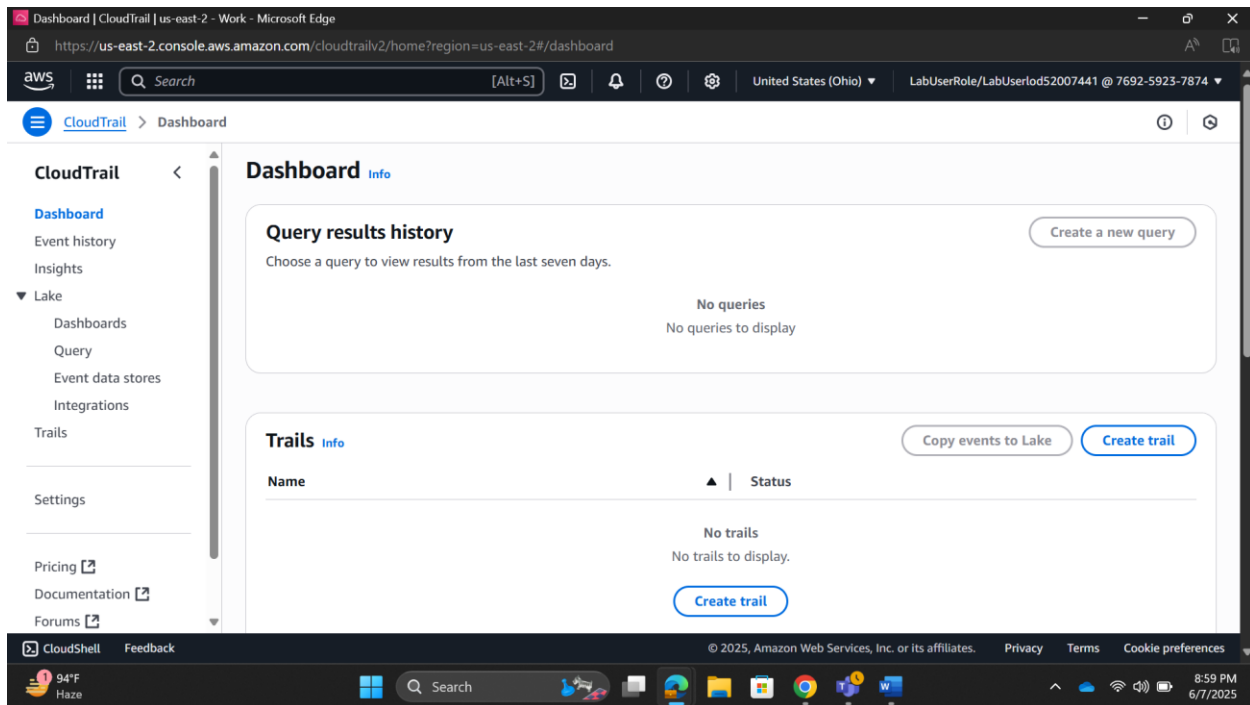


Lab 4 Manage AWS Logs by Using CloudTrail

In this challenge, you will manage AWS logs by using CloudTrail. First, you will enable CloudTrail logging. Next, you will generate log entries. Finally, you will examine and download the logs, and then you will review the event history. Note: Once you begin the challenge, you will not be able to pause, save, or return to your challenge. Please ensure that you have set aside enough time to complete the challenge before you start.

Task 1: Enable CloudTrail logging

CloudTrail -> Dashboard -> trails -> create trail



Create trail | CloudTrail | us-east-2 - Work - Microsoft Edge

https://us-east-2.console.aws.amazon.com/cloudtrailv2/home?region=us-east-2#/create

aws Search [Alt+S] United States (Ohio) LabUserRole/LabUserIod52007441 @ 7692-5923-7874

CloudTrail > Dashboard > Create trail

Step 1
● Choose trail attributes
○ Step 2
○ Choose log events
○ Step 3
○ Review and create

Choose trail attributes

General details

A trail created in the console is a multi-region trail. [Learn more](#)

Trail name
Enter a display name for your trail.

LabTrail-52007441

3-128 characters. Only letters, numbers, periods, underscores, and dashes are allowed.

☐ Enable for all accounts in my organization
To review accounts in your organization, open AWS Organizations. [See all accounts](#)

Storage location [Info](#)

☒ Create new S3 bucket
Create a bucket to store logs for the trail.

☐ Use existing S3 bucket
Choose an existing bucket to store logs for this trail.

Trail log bucket and folder
Enter a new S3 bucket name and folder (prefix) to store your logs. Bucket names must be globally unique.

labtrailbucket-52007441

Logs will be stored in labtrailbucket-52007441/AWSLogs/769259237874

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Create trail | CloudTrail | us-east-2 - Work - Microsoft Edge

https://us-east-2.console.aws.amazon.com/cloudtrailv2/home?region=us-east-2#/create

aws Search [Alt+S] United States (Ohio) LabUserRole/LabUserIod52007441 @ 7692-5923-7874

CloudTrail > Dashboard > Create trail

Log file SSE-KMS encryption [Info](#)

☐ Enabled

▼ **Additional settings**

Log file validation [Info](#)

☒ Enabled

SNS notification delivery [Info](#)

☐ Enabled

CloudWatch Logs - optional
Configure CloudWatch Logs to monitor your trail logs and notify you when specific activity occurs. Standard CloudWatch and CloudWatch Logs charges apply. [Learn more](#)

CloudWatch Logs [Info](#)

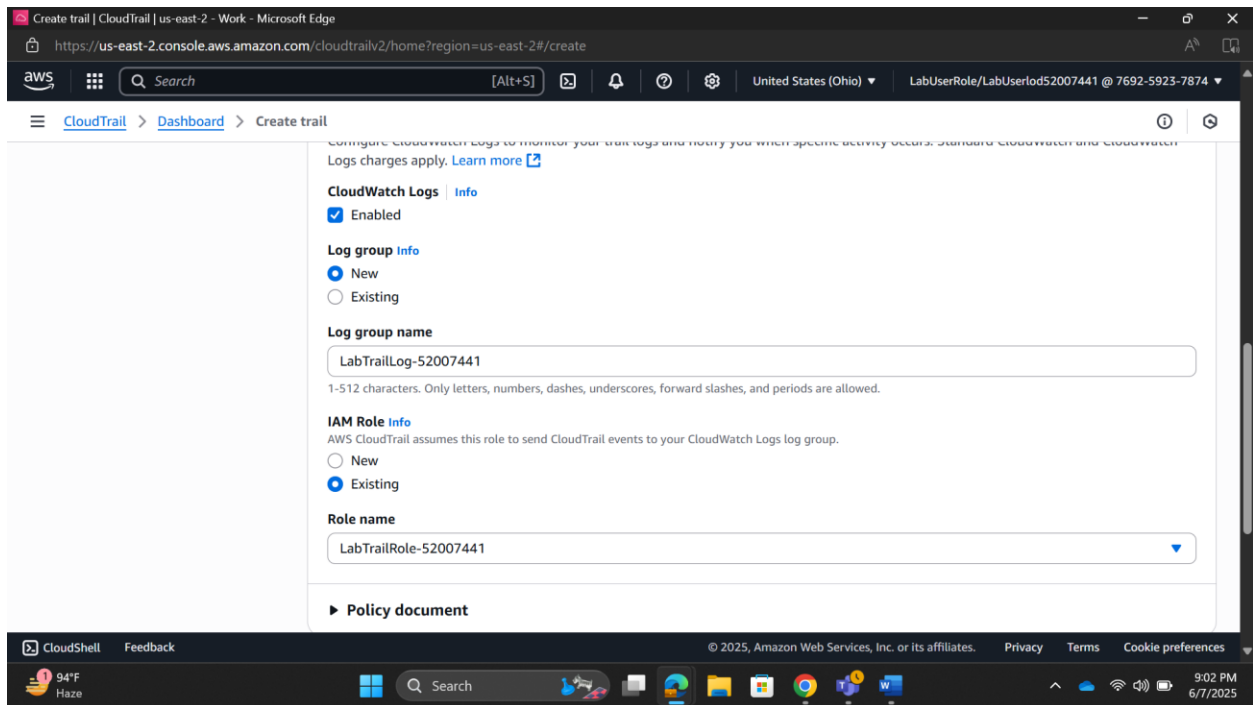
☒ Enabled

Log group [Info](#)

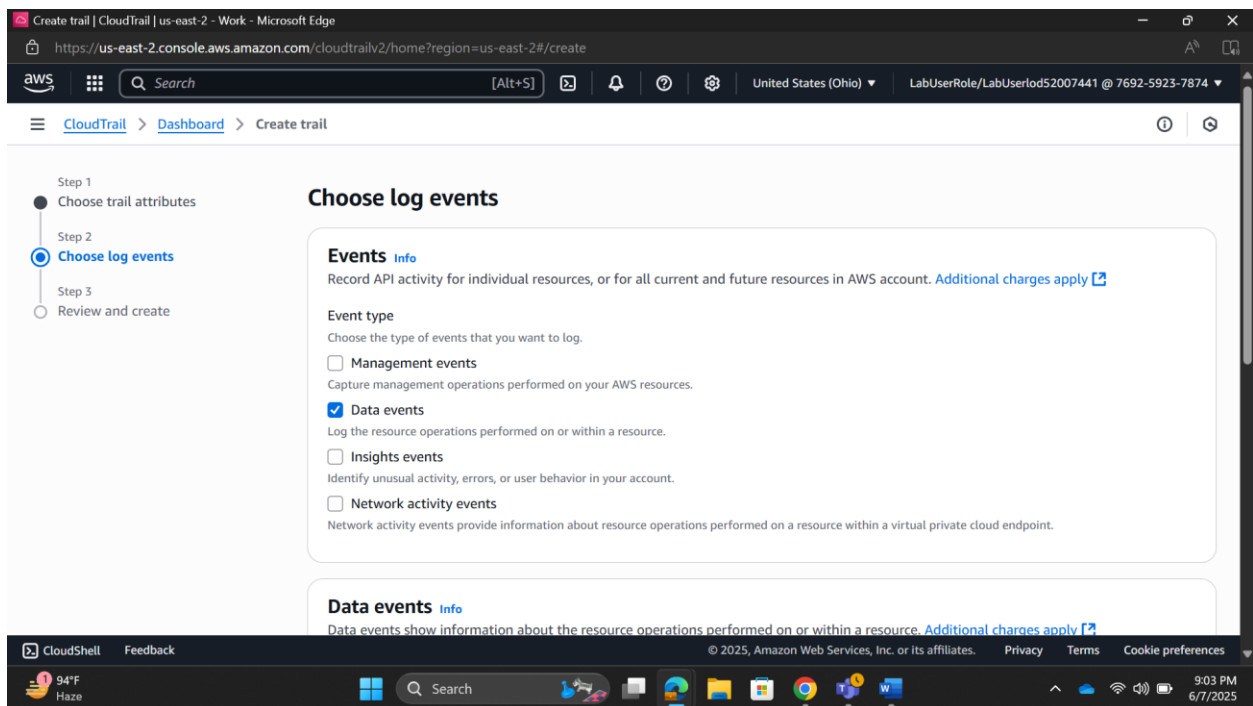
☒ New

☐ Existing

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Click next



Create trail | CloudTrail | us-east-2 - Work - Microsoft Edge

https://us-east-2.console.aws.amazon.com/cloudtrailv2/home?region=us-east-2#/create

aws Search [Alt+S] United States (Ohio) LabUserRole/LabUserIod52007441 @ 7692-5923-7874

CloudTrail > Dashboard > Create trail

▼ Data event: S3 Remove

Resource type
Choose the resource type for which you want to log data events.
S3 ▼

Log selector template
Log all events ▼

Selector name - optional
Enter a name
1,000 character limit

► JSON view

Add data event type

Cancel Previous Next

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Create trail | CloudTrail | us-east-2 - Work - Microsoft Edge

https://us-east-2.console.aws.amazon.com/cloudtrailv2/home?region=us-east-2#/create

aws Search [Alt+S] United States (Ohio) LabUserRole/LabUserIod52007441 @ 7692-5923-7874

CloudTrail > Dashboard > Create trail

Data events: S3
Log selector template
Log all events

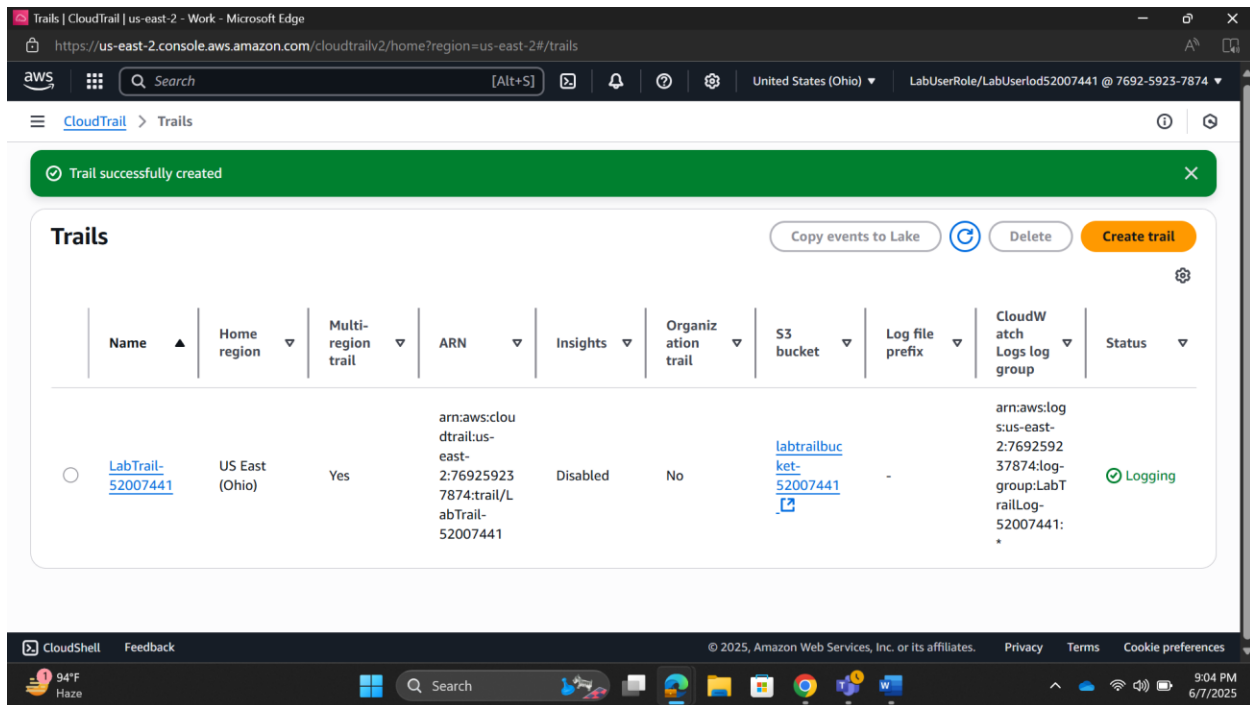
Selector name
--

All events

Insights events
You can only enable CloudTrail Insights on trails that log management events. [Learn more](#)

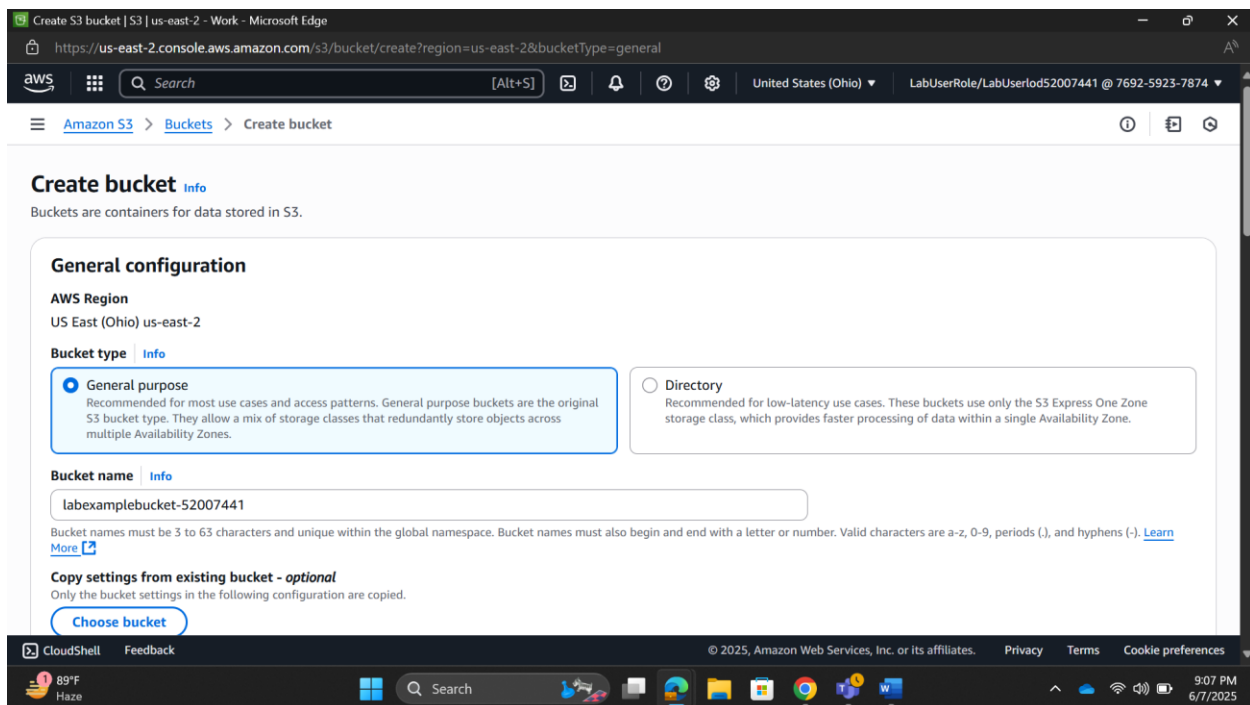
Network activity events
Network activity event collection is not configured for this trail

Cancel Previous Create trail



Task 2: Generate sample management events

S3 -> Create Bucket



Click create bucket

labexamplebucket-52007441 - S3 bucket | S3 | us-east-2 - Work - Microsoft Edge

https://us-east-2.console.aws.amazon.com/s3/buckets/labexamplebucket-52007441?region=us-east-2&bucketType=general&tab=properties

aws

Search

[Alt+S]

United States (Ohio)

LabUserRole/LabUserIod52007441 @ 7692-5923-7874

Amazon S3

Buckets

labexamplebucket-52007441

labexamplebucket-52007441

Info

Objects

Metadata

Properties

Permissions

Metrics

Management

Access Points

Bucket overview

AWS Region

US East (Ohio) us-east-2

Amazon Resource Name (ARN)

arn:aws:s3:::labexamplebucket-52007441

Creation date

June 7, 2025, 21:07:18 (UTC+05:30)

Bucket Versioning

Edit

Versioning is a means of keeping multiple variants of an object in the same bucket. You can use versioning to preserve, retrieve, and restore every version of every object stored in your Amazon S3 bucket. With versioning, you can easily recover from both unintended user actions and application failures. [Learn more](#)

Bucket Versioning

Disabled

Multi-factor authentication (MFA) delete

An additional layer of security that requires multi-factor authentication for changing Bucket Versioning settings and permanently deleting object versions. To modify MFA delete settings, use the AWS CLI, AWS SDK, or the Amazon S3 REST API. [Learn more](#)

Disabled

CloudShell

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Haze

Search

9:08 PM

6/7/2025

Edit versioning - S3 bucket labexamplebucket-52007441 | S3 | us-east-2 - Work - Microsoft Edge

https://us-east-2.console.aws.amazon.com/s3/bucket/labexamplebucket-52007441/property/versioning/edit?region=us-east-2&bucketType=general

aws

Search

[Alt+S]

United States (Ohio)

LabUserRole/LabUserIod52007441 @ 7692-5923-7874

Amazon S3

Buckets

labexamplebucket-52007441

Edit Bucket Versioning

Edit Bucket Versioning

Info

Bucket Versioning

Versioning is a means of keeping multiple variants of an object in the same bucket. You can use versioning to preserve, retrieve, and restore every version of every object stored in your Amazon S3 bucket. With versioning, you can easily recover from both unintended user actions and application failures. [Learn more](#)

Bucket Versioning

☐ Suspend

This suspends the creation of object versions for all operations but preserves any existing object versions.

☒ Enable

After enabling Bucket Versioning, you might need to update your lifecycle rules to manage previous versions of objects.

Multi-factor authentication (MFA) delete

An additional layer of security that requires multi-factor authentication for changing Bucket Versioning settings and permanently deleting object versions. To modify MFA delete settings, use the AWS CLI, AWS SDK, or the Amazon S3 REST API. [Learn more](#)

Disabled

Cancel

Save changes

CloudShell

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Haze

Search

9:08 PM

6/7/2025

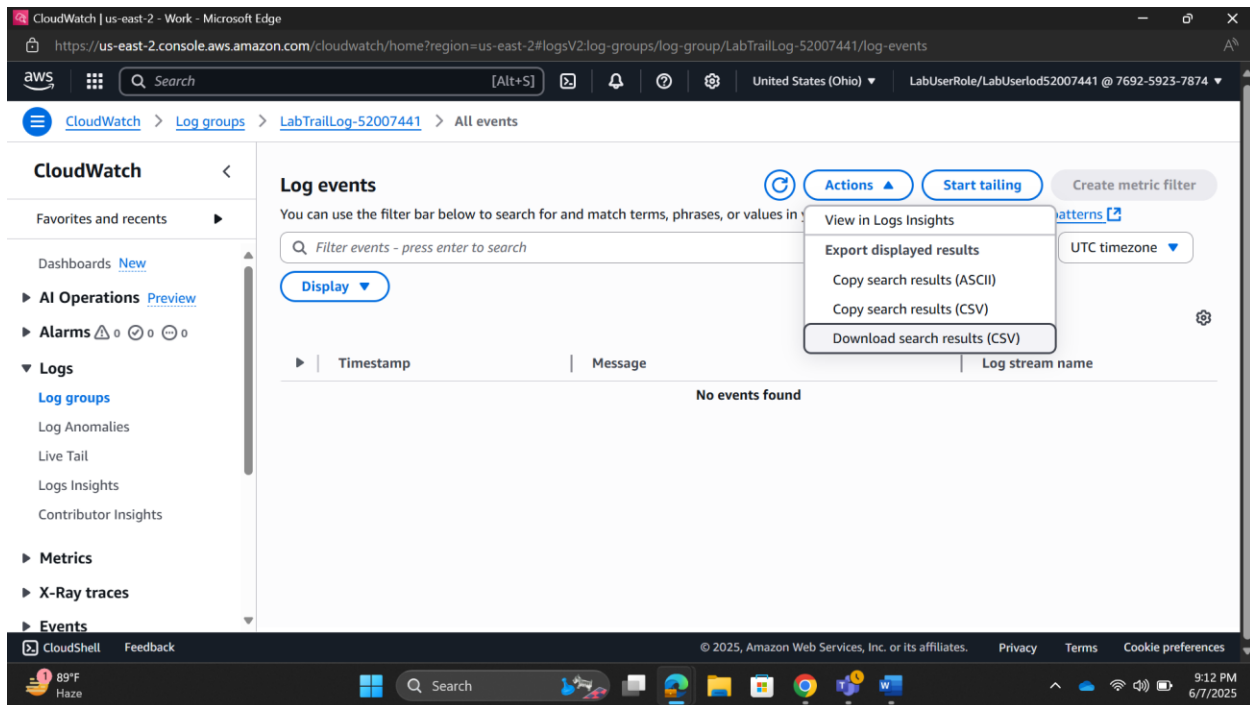
Task 3: View the logs in CloudWatch

CloudWatch -> Logs -> log groups

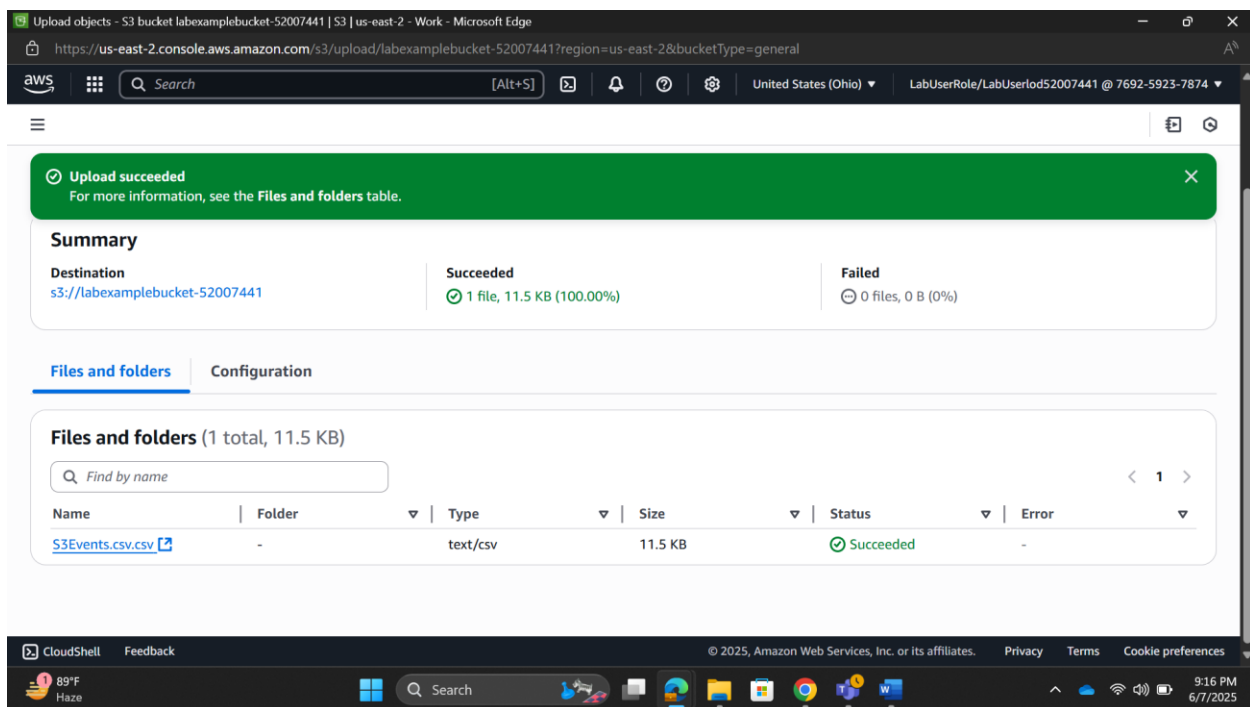
The screenshot shows the AWS CloudWatch console in the 'Log groups' section. The left sidebar contains navigation links for CloudWatch, Favorites and recent, Dashboards, AI Operations, Alarms, Logs, Metrics, X-Ray traces, and Events. The main content area is titled 'Log groups (1)' and includes a search bar, a table of log groups, and buttons for 'Actions', 'View in Logs Insights', 'Start tailing', and 'Create log group'. The table lists one log group: 'LabTrailLog-52007441' with a 'Standard' log class and a 'Configure' link. The bottom of the screen shows the Windows taskbar with the time 9:10 PM on 6/7/2025.

The screenshot shows the AWS CloudWatch console in the 'Log streams' section for the log group 'LabTrailLog-52007441'. The left sidebar is the same as the previous screenshot. The main content area is titled 'Log streams (1)' and includes a search bar, a table of log streams, and buttons for 'Delete', 'Create log stream', and 'Search all log streams'. The table lists one log stream: '769259237874_CloudTrail_us-east-2' with a 'Last event time' of '-'. The top of the main content area shows details for the log group, including 'Creation time' (6 minutes ago), 'Retention' (Never expire), 'Stored bytes' (-), 'Contributor Insights rules' (-), 'KMS key ID' (-), 'Anomaly detection' (Configure), 'Field indexes' (Configure), and 'Transformer' (Configure). The bottom of the screen shows the Windows taskbar with the time 9:11 PM on 6/7/2025.

Click search all log streams



S3 -> open the created bucket -> upload the downloaded file



CloudWatch -> logs -> log groups -> open the log -> search all log streams

CloudWatch | us-east-2 - Work - Microsoft Edge

https://us-east-2.console.aws.amazon.com/cloudwatch/home?region=us-east-2#logsV2:log-groups/log-group/LabTrailLog-52007441/log-events

cloudwatch

United States (Ohio) LabUserRole/LabUserIod52007441 @ 7692-5923-7874

CloudWatch > Log groups > LabTrailLog-52007441 > All events

CloudWatch

Favorites and recents

Dashboards

AI Operations [Preview](#)

Alarms [△](#) [○](#) [○](#) [○](#)

▼ Logs

Log groups

Log Anomalies

Live Tail

Logs Insights

Contributor Insights

Metrics

X-Ray traces

Events

Log events

You can use the filter bar below to search for and match terms, phrases, or values in your log events. [Learn more about filter patterns](#)

Q {\$eventName = "PutBucketVersioning"} X 1m 1h UTC timezone

Display

Timestamp	Message	Log stream name
2025-06-07T15:42:29.086Z	{"eventVersion":"1.11","userIdentity":{"type":"Assume...	769259237874_CloudTrail_us-east-2_2
2025-06-07T15:42:29.086Z	{"eventVersion":"1.11","userIdentity":{"type":"Asser...	769259237874_CloudTrail_us-east-2_2
2025-06-07T15:42:31.778Z	{"eventVersion":"1.11","userIdentity":{"type":"Assume...	769259237874_CloudTrail_us-east-2_3
2025-06-07T15:43:16.675Z	{"eventVersion":"1.11","userIdentity":{"type":"Assume...	769259237874_CloudTrail_us-east-2_4
2025-06-07T15:43:16.675Z	{"eventVersion":"1.11","userIdentity":{"type":"Asser...	769259237874_CloudTrail_us-east-2_4
2025-06-07T15:43:16.675Z	{"eventVersion":"1.11","userIdentity":{"type":"Asser...	769259237874_CloudTrail_us-east-2_4

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89°F Haze 9:19 PM 6/7/2025

Enter filter, click enter

Task 4: View the event history in CloudTrail

CloudTrail -> event history -> PutBucketVersioning event

Event history | CloudTrail | us-east-2 - Work - Microsoft Edge

https://us-east-2.console.aws.amazon.com/cloudtrailv2/home?region=us-east-2#/events?ReadOnly=false

Search [Alt+S]

United States (Ohio) LabUserRole/LabUserIod52014648 @ 1015-0574-7733

CloudTrail > Event history

CloudTrail

Dashboard

Event history

Insights

▼ Lake

Dashboards

Query

Event data stores

Integrations

Trails

Settings

Pricing [🔗](#)

Documentation [🔗](#)

Forums [🔗](#)

Event history (50+) Info

Event history shows you the last 90 days of management events.

Lookup attributes

Read-only Q false X Filter by date and time Clear filter < 1 2 ... >

☐	Event name	Event time	User name	Event source	Resource type
☐	CreateLogStream	June 08, 2025, 12:28:06 (UTC+0...	CLOUDWATCH_LO...	logs.amazonaws.com	-
☐	CreateLogStream	June 08, 2025, 12:22:06 (UTC+0...	CLOUDWATCH_LO...	logs.amazonaws.com	-
☐	PutBucketVersioning	June 08, 2025, 12:17:12 (UTC+0...	LabUserIod52014...	s3.amazonaws.com	AWS::S3::Bucket
☐	PutBucketEncryption	June 08, 2025, 12:16:47 (UTC+0...	LabUserIod52014...	s3.amazonaws.com	AWS::S3::Bucket
☐	CreateBucket	June 08, 2025, 12:16:46 (UTC+0...	LabUserIod52014...	s3.amazonaws.com	AWS::S3::Bucket
☐	UpdateTrail	June 08, 2025, 12:14:15 (UTC+0...	LabUserIod52014...	cloudtrail.amazonaws.c om	AWS::CloudTrail::Tr

0 / 5 events selected

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100°F Haze 12:40 PM 6/8/2025

Event details - CloudTrail - Work - Microsoft Edge

https://us-east-2.console.aws.amazon.com/cloudtrailv2/home?region=us-east-2#/events/1bb4dc63-bb2b-4b7c-b071-119ce3ddee58

aws Search [Alt+S] United States (Ohio) LabUserRole/LabUserIod52014648 @ 1015-0574-7733

CloudTrail > Event history > PutBucketVersioning

PutBucketVersioning Info

Details Info

Event time June 08, 2025, 12:17:12 (UTC+05:30)	AWS access key ASIARPIRWZMK2QSE3Q4P	AWS region us-east-2
User name LabUserIod52014648	Source IP address 103.240.233.161	Error code -
Event name PutBucketVersioning	Event ID 1bb4dc63-bb2b-4b7c-b071-119ce3ddee58	Read-only false
Event source s3.amazonaws.com	Request ID MWD6TGAMV6QA0EXX	

Resources referenced (1) Info

Resource type	Resource name	AWS Config resource timeline
AWS::S3::Bucket	labexamplebucket-52014648	Enable AWS Config resource recording

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100°F Haze 12:41 PM 6/8/2025

Copy the account Id and enter in field

labclient.labondemand.com/Instructions/ExamResult/b7824cc5-96ac-44d2-a217-fa4e74718d26 - Work - Microsoft Edge

https://labclient.labondemand.com/Instructions/ExamResult/b7824cc5-96ac-44d2-a217-fa4e74718d26

Congratulations!

Print

Congratulations on your success!



Areas where you excelled

- **Enable CloudTrail logging**
You completed this area with great success.
- **Generate sample management events**
You showed your skills in this area and finished with success.
- **View the logs in CloudWatch**
You were really strong in this area.
- **View the event history in CloudTrail**

93°F Haze 9:33 AM 6/13/2025

You are done with the lab!!!